University of Illinois
ECE 310 **Profs. Do, Shomorony, Snyder**

Midterm Exam 2

7:00-9:00pm, Wednesday, November 6, 2024

Name: _____

Section: 9:00 AM 12:00 PM 3:00 PM

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	8	
3	12	
4	12	
5	22	
6	8	
7	6	
8	8	
9	12	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than two handwritten two-sided sheets of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

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(12 Pts.)

1. Select **True** or **False** to each of the following statements. Please explain your reasoning in **no more than two sentences.**

   (a) The z-transform of $x[n] = 2^n u[n]$ is given by $X(z) = \frac{1}{1-2z^{-1}},\ |z| > 2$. Therefore, the DTFT of $x[n]$ is given by $X_d(\omega) = \frac{1}{1-2e^{-j\omega}}$. **True/<u>False</u>**

   The given signal $x[n]$ is not absolutely summable; therefore, the DTFT $X_d(\omega)$ does not exist.

   (b) The DTFT $X_d(\omega)$ of a finite-length signal $x[n]$ is 2π-periodic. **<u>True</u>/False**

   The DTFT of any signal, if it exists, is always 2π-periodic.

   (c) A discrete-time signal $x[n] = 2\delta[n-5]$ is passed to an ideal D/A converter with sampling period T to recover $x_a(t)$. The recovered signal is given by $x_a(t) = 2\delta(t-5T)$. **True/<u>False</u>**

   An ideal D/A converter results in sinc interpolation in the time-domain; thus we should have $x_a(t) = 2\text{sinc}\left(\frac{\pi}{T}(t-5T)\right)$.

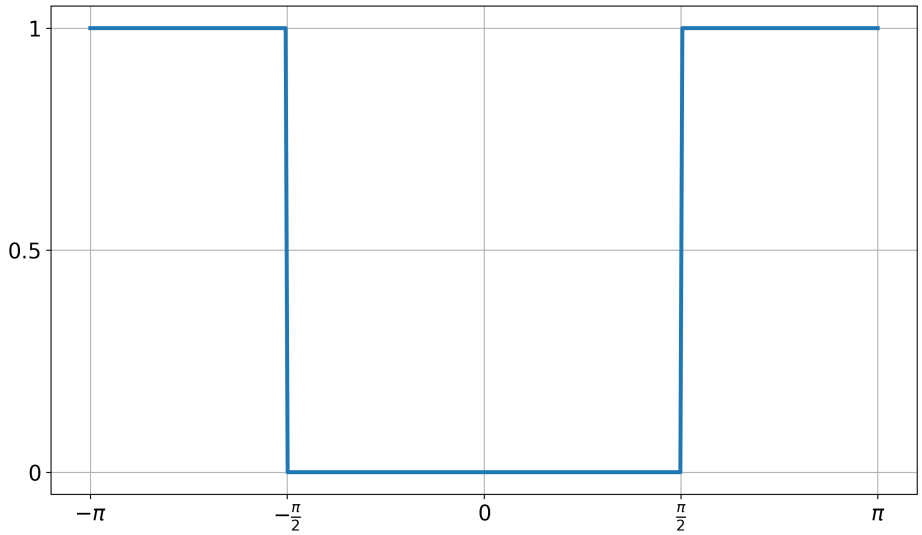
   (d) A length-N signal $x[n]$ with DTFT $X_d(\omega)$ and DFT $X[k]$ is zero-padded with N zeros to length-$2N$ to form $y[n]$. Then, $X[k] = Y[2k]$ and $X_d(\omega) = Y_d(\omega)$. **<u>True</u>/False**

   Zero-padding in this case will double the number of values in the DFT and thus reduce the frequency spacing by half in the DFT. This means $X[k] = Y[2k]$ while the DTFT is not affected by zero-padding.

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(8 Pts.)

2. The below figure shows the DTFT $X_d(\omega)$ of the signal $x[n]$ for $-\pi \leq \omega \leq \pi$. Determine $x[n]$. For full credit, your expression for $x[n]$ must be real-valued.



$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_d(\omega)e^{j\omega n}d\omega$$
$$= \frac{1}{2\pi}\int_{-\pi}^{-\frac{\pi}{2}} e^{j\omega n}d\omega + \frac{1}{2\pi}\int_{\frac{\pi}{2}}^{\pi} e^{j\omega n}d\omega$$
$$= \frac{e^{-j\frac{\pi}{2}n} - e^{-j\pi n} + e^{j\pi n} - e^{j\frac{\pi}{2}n}}{2\pi jn}$$
$$= \frac{e^{j\pi n} - e^{-j\pi n}}{2\pi jn} - \frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{2\pi jn}$$
$$= \frac{\sin(\pi n)}{\pi n} - \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}$$
$$= \delta[n] - \frac{1}{2}\text{sinc}\left(\frac{\pi}{2}n\right)$$

$$x[n] = \frac{1}{2\pi}\int_{0}^{2\pi} X_d(\omega)e^{j\omega n}d\omega$$
$$= \frac{1}{2\pi}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} e^{j\omega n}d\omega$$
$$= \frac{e^{j\frac{3\pi}{2}n} - e^{j\frac{\pi}{2}n}}{2\pi jn}$$
$$= e^{j\pi n}\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{2\pi jn}$$
$$= (-1)^n\frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}$$
$$= (-1)^n\frac{1}{2}\text{sinc}\left(\frac{\pi}{2}n\right)$$

$$X_d(\omega) = Y_d\left(\omega - \frac{3\pi}{4}\right) + Y_d\left(\omega + \frac{3\pi}{4}\right)$$
$$Y_d(\omega) = \text{rect}\left(\frac{\omega}{\pi/2}\right)$$
$$x[n] = 2\cos\left(\frac{3\pi}{4}n\right)y[n]$$
$$y[n] = \frac{1}{4}\text{sinc}\left(\frac{\pi}{4}n\right)$$
$$x[n] = \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right)\text{sinc}\left(\frac{\pi}{4}n\right)$$

$x[n]$ = Any of the above answers

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(12 Pts.)

3. Suppose we have an LTI system with impulse response

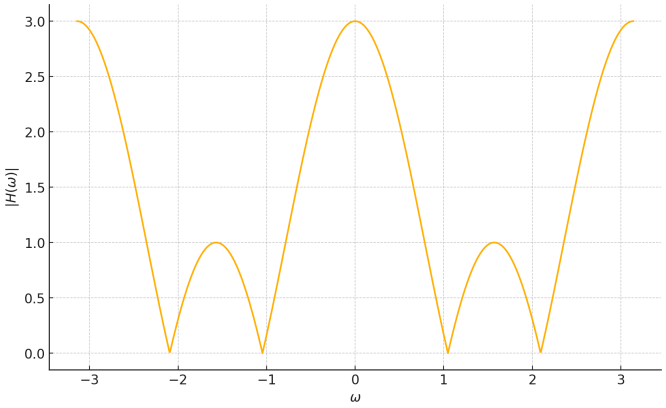
$$h[n] = \delta[n+2] + \delta[n] + \delta[n-2].$$

(a) Find the frequency response $H_d(\omega)$ of this system for the values of ω specified below:

$$\begin{aligned} H_d(\omega) &= e^{j2\omega} + 1 + e^{-j2\omega)} \\ &= 2\cos(2\omega) + 1. \end{aligned}$$

$H_d(0) = 3$	$H_d(\frac{\pi}{2}) = -1$	$H_d(\pi) = 3$

(b) Plot the magnitude response $|H_d(\omega)|$ for $-\pi \leq \omega \leq \pi$. Make sure to carefully label your plot.



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(12 Pts.)

4. The frequency response of an LTI system is given by

$$H_d(\omega) = |\omega| e^{-j\pi \sin(\omega)}, \quad \text{for } |\omega| \leq \pi.$$

(a) Is the impulse response $h[n]$ of this system real-valued? Justify your reasoning.

$$\begin{aligned} H_d(-\omega) &= |-\omega| e^{-j\pi \sin(-\omega)} \\ &= |\omega| e^{j\pi \sin(\omega)} \\ H_d^*(\omega) &= |\omega| e^{j\pi \sin(\omega)} \end{aligned}$$

The above expressions are equal; therefore, $h[n]$ is real-valued by Hermitian symmetry.

(b) Determine the ouptut $y[n]$ when the input is $x[n] = 3 + \cos\left(\frac{\pi n}{6}\right) + j^n$.

$$\begin{aligned} H_d(0) &= 0 \\ |H_d(\pi/6)| &= \frac{\pi}{6}, \ \angle H_d(\pi/6) = -\frac{\pi}{2} \\ H_d(\pi/2) &= -\frac{\pi}{2} \end{aligned}$$

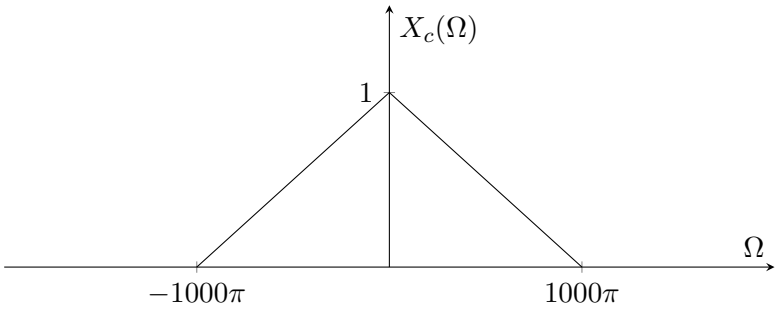
Applying these values of the frequency response, we find:

$$\begin{aligned} y[n] &= \frac{\pi}{6} \cos\left(\frac{\pi}{6} n - \frac{\pi}{2}\right) - \frac{\pi}{2} j^n \\ y[n] &= \frac{\pi}{6} \sin\left(\frac{\pi}{6} n\right) - \frac{\pi}{2} j^n \\ y[n] &= \frac{\pi}{6} \cos\left(\frac{\pi}{6} n - \frac{\pi}{2}\right) - \frac{\pi}{2} e^{j\frac{\pi}{2} n} \\ y[n] &= \frac{\pi}{6} \sin\left(\frac{\pi}{6} n\right) - \frac{\pi}{2} e^{j\frac{\pi}{2} n} \end{aligned}$$

$y[n]$ = Any of the above answers

(22 Pts.)

5. The continuous-time signal $x_c(t)$ has the continuous-time Fourier transform (CTFT) shown in the figure below. The signal $x_c(t)$ is sampled with sampling interval T to get the discrete-time signal $x[n] = x_c(nT)$. Then, the discrete-time signal $x[n]$ is passed through an ideal digital-to-analog (D/A) converter with interval T to get the continuous-time signal $y_c(t)$.



(a) **(2 Pts.)** Find the Nyquist sampling period T_N (of the Nyquist sampling frequency) of the given signal $x_c(t)$.

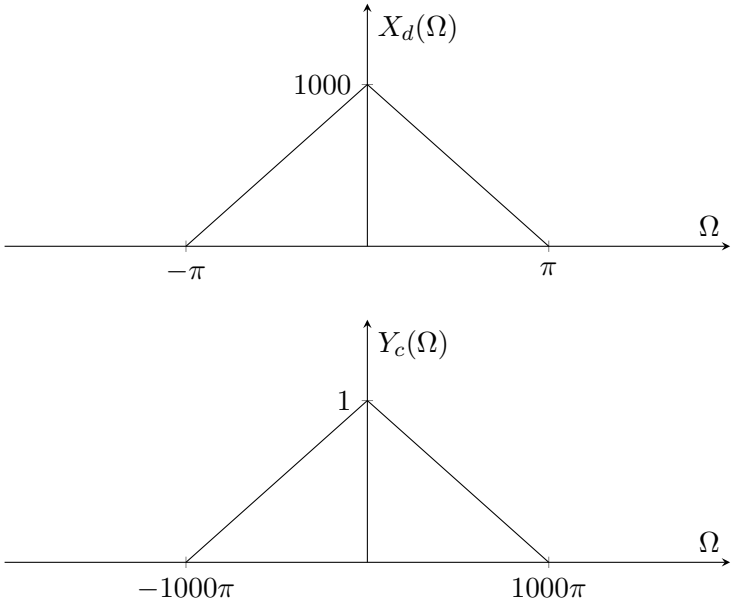
$$T_N = \frac{1}{1000} \text{ s} = 1 \text{ ms}$$

(b) **(12 Pts.)** Sketch $X_d(\omega)$ (the DTFT of $x[n]$) and $Y_c(\Omega)$ (the CTFT of $y_c(t)$) for each of the following sampling intervals: $T = T_N$, $T = 2T_N$. Clearly label all of the axes in your sketches. Plots will be added later, in the meantime:

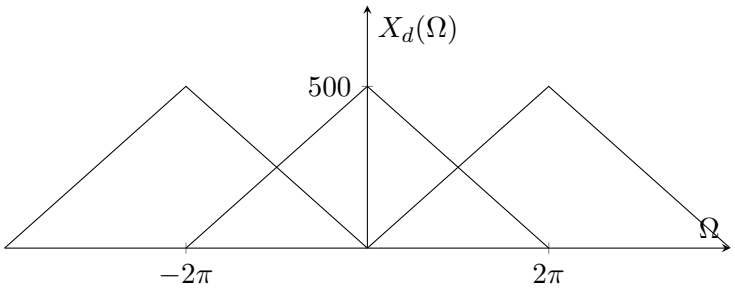
i. $T = T_N$: The spectral copies should not overlap but touch at their edges.

ii. $T = 2T_N$: The spectral copies will fully overlap and lead to a constant spectrum.

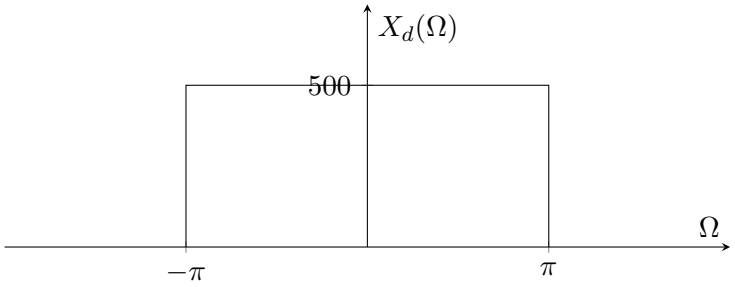
For $T = T_N$ case:



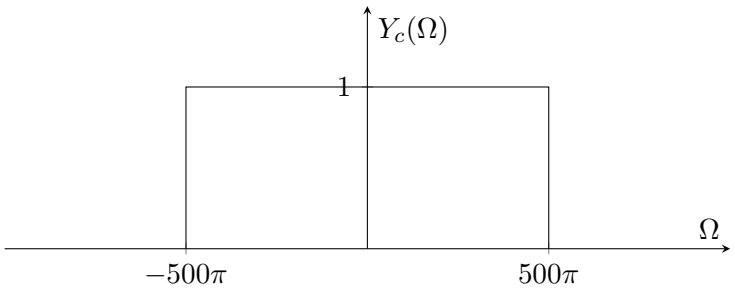
For $T = 2T_N$ case:



After summing the overlapping, in one period we get the DTFT as follows:



Plot of CTFT of $y_c(t)$ is:



(c) **(8 Pts.)** Suppose we would like to filter out frequencies $|\Omega| > 500\pi$ in $x_c(t)$ by applying an ideal low-pass filter (LPF) to discrete-time signal $x[n]$ before passing through the ideal D/A converter. What is the largest sampling interval T for the ideal A/D and D/A converters that we can use to accomplish this task (note that some aliasing is allowable here)? What is the cut-off frequency ω_c of the ideal discrete-time LPF for this largest value of T?

The rightmost frequency of the central copy in $X_d(\omega)$ will be located at $1000\pi T$, while the leftmost frequency of the right spectral copy (centered at 2π) will be located at $2\pi - 1000\pi T$. However, we want to avoid aliasing present in the output of the digital filter, thus we need to ensure no aliasing occurs within the LPF passband.

For sampling period T, the necessary digital filter's cutoff frequency will be at $\omega_c = 500\pi T$. Thus, we want to ensure the adjacent spectral copies do not reach inside of this passband. Therefore, we need:

$$500\pi T < 2\pi - 1000\pi T$$
$$T < \frac{1}{750} \text{ s.}$$

Using this largest value of T, we arrive at our answer.

$$\boxed{T = \frac{1}{750} \text{ s}, \ \omega_c = \frac{2\pi}{3}}$$

(8 Pts.)

6. Suppose that we sample a continuous-domain signal $x_c(t)$ (which is not necessary bandlimited) with sampling frequency $f_s = 1000$ Hz to produce $x[n] = x_c(n/f_s)$. Then, $N = 1024$ samples $\{x[n]\}_{n=0}^{N-1}$ are extracted as input to the length-N DFT to produce the output $\{X[k]\}_{k=0}^{N-1}$. Which continuous-domain frequencies of $x_c(t)$ in Hz are captured in the DFT output $X[k]$ for a given index k? Clearly justify your answer. Note that due to possible aliasing, multiple continuous-domain frequencies Ω can map to the same discrete-domain frequency $\omega \in [0, 2\pi]$.

For each index k, the DFT $X[k]$ will evaluate the DTFT $X_d(\omega)$ at the frequency $\omega_k = \frac{2\pi k}{N}$, where N is the length of the signal. However, due to the 2π-periodicity of the DTFT (and correspondingly N-periodicity of the DFT), we know that any frequency content between $[-\pi, \pi]$, correspondingly $[0, 2\pi]$, also appears at all 2π-periodic copies of the DTFT. Therefore, we may say that each index k captures:

$$\begin{aligned}\omega_k &= \frac{2\pi(k + mN)}{N},\ m \in \mathbb{Z} \\ \omega_k &= \Omega_k T \\ \Omega &= 2\pi f\end{aligned}$$

Thefore, for index k and linear frequency f_k:

$$\begin{aligned}\Omega_k &= \frac{2\pi(k + 1024m)}{1024 \cdot \frac{1}{1000}} \\ &= \frac{2000\pi k}{1024} + 2000\pi m\end{aligned}$$

$$\boxed{f_k = \frac{1000k}{1024} + 1000m \text{ Hz},\ m \in \mathbb{Z}}$$

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(6 Pts.)

7. Suppose we have a length-N discrete-time signal $x[n]$ with DTFT $X_d(\omega)$ and DFT $X[k]$. We know that $X[k]$ for this particular $x[n]$ is able to capture the values $X_d\left(\frac{2\pi}{5}\right)$ and $X_d\left(\frac{\pi}{3}\right)$ in $X_d(\omega)$. Determine the smallest possible value of N.

For length N, a given DFT will be able to capture frequencies $\omega_k = \frac{2\pi k}{N}$ from the DTFT. We need to determine a value N that will make $\frac{2\pi}{5}$ and $\frac{\pi}{3}$ accessible in the above ω_k expression.

We may obtain $\frac{2\pi}{5}$ if N is any multiple of 5, while we may obtain $\frac{\pi}{3}$ if N is any multiple of 6. Therefore, the first multiple of both 5 and 6 is 30, and the smallest possible value of N is 30.

$N = 30$

(8 Pts.)

8. A length-5 signal $\{x[n]\}_{n=0}^{4} = \{3, 0, -1, 2, 4\}$ has DFT $\{X[k]\}_{k=0}^{4} = \{X_0, X_1, X_2, X_3, X_4\}$. Consider the signal $\{y[n]\}_{n=0}^{4} = \{0, 3, 4, 2, -1\}$. Determine $\{Y[k]\}_{k=0}^{4}$ in terms of X_0, X_1, X_2, X_3, X_4.

The given $y[n]$ is the result of circularly time-reversing $x[n]$ and then circularly shifting right by 1. (Alternatively, circular shift left by 4; or, circular shift left by 1, then circular time-reversal will arrive at the same answer too.) Thus, applying circular time reversal and shifting properties:

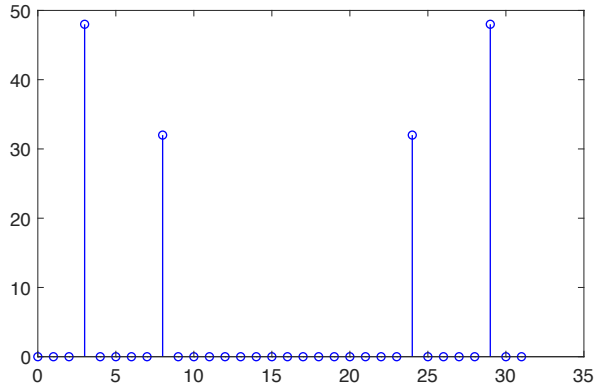
$$y[n] = x[\langle -(n-1)\rangle_5]$$
$$Y[k] = X[\langle -k\rangle_5]e^{-j\frac{2\pi k}{5}}$$

$Y[k] = \{X_0, X_4e^{-j\frac{2\pi}{5}}, X_3e^{-j\frac{4\pi}{5}}, X_2e^{-j\frac{6\pi}{5}}, X_1e^{-j\frac{8\pi}{5}}\}$

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(12 Pts.)

9. A continuous-time signal $x_c(t) = A_1 \cos(\Omega_1 t) + A_2 \cos(\Omega_2 t)$ is sampled at a rate of 64 Hz for 0.5 seconds to produce a discrete-time signal $\{x[n]\}_{n=0}^{31}$. The DFT of $\{x[n]\}_{n=0}^{31}$ was computed and is shown below.



Assuming that A_1 and A_2 are integers and $A_1 > A_2$, find the values of A_1, A_2, Ω_1 and Ω_2.

For index k and length N, we have the DFT evaluated the DTFT at frequency $\omega_k = \frac{2\pi k}{N}$. We may also relate digital frequencies to analog frequencies via $\omega = \Omega T$. For the provided values, we may then find

$$\Omega_1 = \frac{\omega_1}{T} = \frac{2\pi \cdot 3}{32 \cdot \frac{1}{64}} = 12\pi \text{ rad/s.}$$
$$\Omega_2 = \frac{\omega_2}{T} = \frac{2\pi \cdot 8}{32 \cdot \frac{1}{64}} = 32\pi \text{ rad/s.}$$

Next, each spectral component is represented by a periodic sinc function with peak value $\frac{AN}{2}$ for amplitude A and length N. Thus, we know that the respective heights of each spectral component in the DFT plot will be multiples of $\frac{N}{2} = 16$ since A_1 and A_2 are integers. Observing the plot, we may conclude that $A_1 = 3$ and $A_2 = 2$

$A_1 = 3$	$A_2 = 2$	$\Omega_1 = 12\pi$	$\Omega_2 = 32\pi$

Scratch paper. Do not remove from the exam!